

Usually with definite integrals we assume $f(x)$ doesn't have an infinite discontinuity and our interval of integration is discrete.

If we take away these assumptions, we can still integrate carefully.

We separate these into two types of improper integrals

Type 1: Infinite intervals

i.e. $\int_0^{\infty} f(x) dx$

Type 2: Discontinuous Integrand

i.e. $f(x)$ has infinite discontinuity

Type 1: Infinite Intervals

Consider $\int_1^{\infty} \frac{1}{x^2} dx$. Evaluating for an infinite number is not reasonable, so we will use limit notation.

$$\lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx$$

Now, this is a finite integral for each choice of t so

$$\lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^2} dx = \lim_{t \rightarrow \infty} \left(1 - \frac{1}{t}\right) = 1$$

So, the area under the curve $\frac{1}{x^2}$ on $[1, \infty)$ is finite even though the arc-length is infinite.

Def:

1) If $\int_a^t f(x) dx$ exists for every $t \geq a$, then

$$\int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

given the limit exists and is finite.

2) If $\int_t^b f(x) dx$ exists for every $t \leq b$ then

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

given the limit exists and is finite.

The integrals $\int_a^{\infty} f(x) dx$ and $\int_{-\infty}^b f(x) dx$ are said to be convergent if their corresponding limit exists and is finite, and are divergent otherwise.

3) If both $\int_a^{\infty} f(x) dx$ and $\int_{-\infty}^a f(x) dx$ are convergent, then

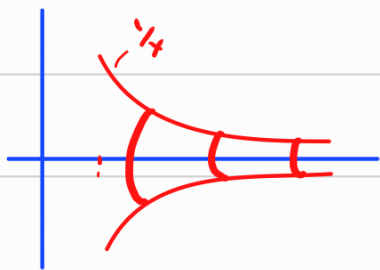
$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^a f(x) dx + \int_a^{\infty} f(x) dx$$

Ex: $\int_1^{\infty} \frac{1}{x} dx$

$$\begin{aligned} \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x} dx &= \lim_{t \rightarrow \infty} \ln|x| \Big|_1^t \\ &= \lim_{t \rightarrow \infty} \ln(t) = \infty \end{aligned}$$

$$\int_1^{\infty} \frac{1}{x} dx \text{ is } \underline{\text{divergent}}$$

Ex: If you recall Solids of revolution



$$\pi \int_1^{\infty} \left(\frac{1}{x}\right)^2 dx$$
$$= \pi$$

So this Shape has finite volume.

However, we will show later the surface area is infinite.

Ex:

$$\int_{-\infty}^0 x e^x dx$$

$$\lim_{t \rightarrow -\infty} \int_t^0 x e^x dx$$

$$u = x \quad dv = e^x dx$$
$$du = dx \quad v = e^x$$

$$= \lim_{t \rightarrow -\infty} x e^x \Big|_t^0 - \int_t^0 e^x dx$$

$$= \lim_{t \rightarrow -\infty} x e^x - \lim_{t \rightarrow -\infty} \int_t^0 e^x dx$$

$$\lim_{t \rightarrow -\infty} \frac{x}{\frac{1}{e^x}} - \lim_{t \rightarrow -\infty} e^x \Big|_t^0$$

L'H

$$= \lim_{t \rightarrow -\infty} \frac{1}{-e^x} - 1$$

$$= 0 - 1 = -1$$

So, $\int x e^x dx$ Converges to -1

Ex: $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$, by convenience pick 0 for separate integral

$$= \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2}$$

$$\int_{-\infty}^0 \frac{dx}{1+x^2} = \lim_{t \rightarrow -\infty} \int_t^0 \frac{dx}{1+x^2} = \lim_{t \rightarrow -\infty} \arctan(x) \Big|_t^0$$

$$= \lim_{t \rightarrow -\infty} -\arctan(t) = \frac{\pi}{2}$$

$$\int_0^{\infty} \frac{dx}{1+x^2} = \lim_{t \rightarrow \infty} \int_0^t \frac{dx}{1+x^2} = \lim_{t \rightarrow \infty} \arctan(t) = \frac{\pi}{2}$$

S-

$$\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \int_{-\infty}^0 \frac{dx}{1+x^2} + \int_0^{\infty} \frac{dx}{1+x^2} = \boxed{\pi}$$

An important integral to observe is the

general

$$\int_1^{\infty} \frac{1}{x^p} dx \quad \text{for } p > 0$$

• If $p=1$, $\int_1^{\infty} \frac{1}{x} dx$ is divergent (see above)

$$\text{• If } p > 1, \quad \int_1^{\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \int_1^t \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \frac{1}{1-p} x^{-p+1} \Big|_1^t = \frac{1}{1-p} \left(\lim_{t \rightarrow \infty} t^{-p+1} - 1 \right)$$

Since $p > 1$, $-p+1 < 0$ so $\lim_{t \rightarrow \infty} t^{-p+1} = 0$ and the integral converges.

• If $0 < p < 1$,

$$\int_1^{\infty} \frac{1}{x^p} dx = \lim_{t \rightarrow \infty} \frac{1}{-p+1} \left(t^{-p+1} - 1 \right)$$

but $-p+1 > 0$ so $\lim_{t \rightarrow \infty} t^{-p+1} = \infty$

and the integral diverges.

Key Point:

$\int_1^{\infty} \frac{dx}{x^p}$ is convergent for $p > 1$, and divergent for $p \leq 1$.

Type 2 Improper Integrals

Say $f(x)$ has an infinite discontinuity at b ?

How do we evaluate $\int_a^b f(x) dx$?

Def:

- 1) If f is continuous on $[a, b)$ and discontinuous at b , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow b^-} \int_a^t f(x) dx$$

if the limit exists and is finite

- 2) If f is continuous on $(a, b]$ but discontinuous at a , then

$$\int_a^b f(x) dx = \lim_{t \rightarrow a^+} \int_t^b f(x) dx$$

if the limit exists and is finite.

The improper integral $\int_a^b f(x) dx$ is convergent if the corresponding limit is finite, it is said to be divergent otherwise.

- 3) If $f(x)$ is discontinuous at $c \in (a, b)$ then

$$\int_a^b f(x) dx = \int_a^c f(x) + \int_c^b f(x) dx \quad \text{if each integral is convergent}$$

$$\underline{\text{Ex:}} \quad \int_2^5 \frac{dx}{\sqrt{x-2}} = \lim_{t \rightarrow 2^+} \int_t^5 \frac{dx}{\sqrt{x-2}} = \lim_{t \rightarrow 2^+} 2\sqrt{x-2} \Big|_t^5 \\ = \lim_{t \rightarrow 2^+} (2\sqrt{3} - 2\sqrt{t-2}) \\ = 2\sqrt{3}$$

$$\text{So} \quad \int_2^5 \frac{dx}{\sqrt{x-2}} \text{ Converges to } 2\sqrt{3}$$

$$\underline{\text{Ex:}} \quad \int_0^{\pi/2} \sec x \, dx = \lim_{t \rightarrow \pi/2^-} \int_0^t \sec x \, dx$$

$$= \lim_{t \rightarrow \pi/2^-} \left[\ln |\tan x + \sec x| \right]_0^t \\ = \lim_{t \rightarrow \pi/2^-} \ln |\tan t + \sec t| - \ln |\tan 0 + \sec 0| \\ = \lim_{t \rightarrow \pi/2^-} \ln |\tan t + \sec t| - \ln(1) \\ = \infty \quad \text{since } \lim_{t \rightarrow \pi/2^-} \tan(t) = \infty \text{ and } \lim_{t \rightarrow \pi/2^-} \sec(t) = \infty$$

$$\text{So, } \int_0^{\pi/2} \sec(x) \, dx \text{ diverges}$$

Ex: $\int_0^3 \frac{dx}{x-1}$

$$= \int_0^1 \frac{dx}{x-1} + \int_1^3 \frac{dx}{x-1}$$

$$= \lim_{t \rightarrow 1^+} \int_0^t \frac{dx}{x-1} + \lim_{t \rightarrow 1^-} \int_t^3 \frac{dx}{x-1}$$

$$\lim_{t \rightarrow 1^+} \int_0^t \frac{dx}{x-1} = \lim_{t \rightarrow 1^+} \ln|x-1| \Big|_0^t = \lim_{t \rightarrow 1^+} \ln|t-1| - \ln|1|$$

$$= \infty$$

Since $\int_0^1 \frac{dx}{x-1}$ diverges

the entire integral

$\int_0^3 \frac{dx}{x-1} \text{ diverges}$

Notice if we did not break this integral up and instead made the mistake

$$\int_0^3 \frac{dx}{x-1} = \ln|x-1| \Big|_0^3 = \ln(2) - \ln(0)$$

$$= \ln(2)$$

is finite even though we showed the integral diverges.

Ex:

$$\int_0^1 \ln x \, dx$$

$$= \lim_{t \rightarrow 0^+} \int_t^1 \ln x \, dx \quad \begin{array}{l} u = \ln x \quad dv = dx \\ du = \frac{1}{x} dx \quad v = x \end{array} = \lim_{t \rightarrow 0^+} x \ln x \Big|_t^1 - \int_t^1 dx$$

$$= \lim_{t \rightarrow 0^+} \frac{\ln(t)}{\frac{1}{t}} - \lim_{t \rightarrow 0^+} x \Big|_t^1$$

$\downarrow$ L'H

$$= \lim_{t \rightarrow 0^+} \frac{y_t}{-1/t_t} - 1$$

$$= \lim_{t \rightarrow 0^+} t - 1$$

$$= -1$$

So, $\int_0^1 \ln x \, dx$ Converges to -1

like the past section, let's look at

If $p=1$

$$\int_0^1 \frac{1}{x} dx = \lim_{t \rightarrow 0^+} -\ln t = \infty, \text{ so it diverges.}$$

If $p \neq 1$

$$\int_0^1 \frac{1}{x^p} dx = \lim_{t \rightarrow 0^+} \int_t^1 \frac{1}{x^p} dx = \frac{1}{-p+1} \left(1 - \lim_{t \rightarrow 0^+} t^{-p+1} \right)$$

If $p > 1$,

$$-p+1 < 0 \text{ so } \lim_{t \rightarrow 0^+} t^{-p+1} = \infty \rightarrow \text{diverge}$$

If $p < 1$

$$-p+1 > 0 \text{ so } \lim_{t \rightarrow 0^+} t^{-p+1} = 0 \rightarrow \text{Converge}$$

Key Point:

$$\int_0^1 \frac{1}{x^p} dx \text{ converges for } p < 1 \text{ and diverges for } p \geq 1$$

Comparison Test

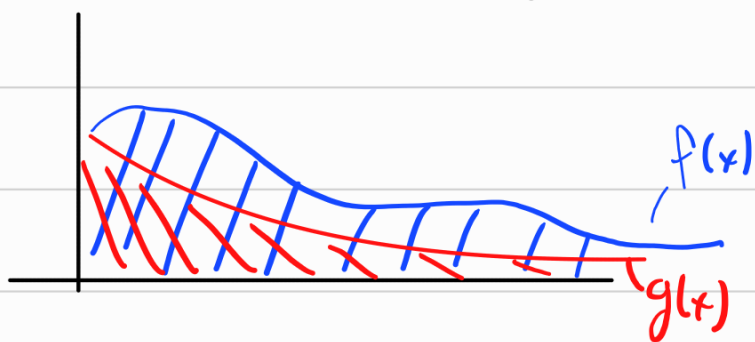
We don't know how to integrate all continuous functions, however we still have methods to show they converge or diverge.

Comparison Test

Suppose f and g are continuous functions with $f(x) \geq g(x) \geq 0$ for $x \geq a$

- a) If $\int_a^{\infty} f(x) dx$ is convergent, $\int_a^{\infty} g(x) dx$ is too.
b) If $\int_a^{\infty} g(x) dx$ is divergent, $\int_a^{\infty} f(x) dx$ is too.

Think of the following picture



• If the area under $g(x)$ is infinite, the area under $f(x)$ is infinite too (as it is more).

• If area under $f(x)$ is finite, Area under $g(x)$ is finite too as it is less.

Ex: $\int_0^{\infty} e^{-x^2} dx$ is convergent.

on $[0, 1)$ $e^{-x} \geq e^{-x^2}$ but this is
a finite area anyways. So, let's look at

$[1, \infty)$
Notice, $\int_1^{\infty} e^{-x} dx = 1$ is convergent, on
 $[1, \infty)$ we have $0 \leq e^{-x^2} \leq e^{-x}$, so

$$\int_1^{\infty} e^{-x^2} dx \leq \int_1^{\infty} e^{-x} dx \quad \text{and } \int_1^{\infty} e^{-x} dx \text{ is convergent}$$

Overall, $\int_0^{\infty} e^{-x^2} dx$ is convergent.